Skin Cancer Classification using Deep Learning and Explainable AI

1. Introduction

Skin cancer is one of the most common diseases worldwide, where early detection plays a critical role in improving survival rates. This project aims to develop a robust deep learning pipeline for **multiclass skin lesion classification** using dermoscopic images, while integrating **Explainable AI (XAI)** techniques to ensure interpretability.

The system leverages **ResNet50** with transfer learning to classify skin lesions into multiple diagnostic categories.

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2. Dataset and Data Integration

2.1 Data Sources

Two major datasets were used:

- HAM10000 dataset
- ISIC Skin Cancer dataset

2.2 Dataset Unification

- Both datasets were merged into a single directory (`unified_dataset`)
- Labels were standardized into common categories:
    - Melanoma
    - Melanocytic nevi
    - Basal cell carcinoma
    - Actinic keratosis
    - Benign keratosis
    - Dermatofibroma
    - Vascular lesion

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3. Exploratory Data Analysis (EDA)

3.1 Class Distribution

- Initial dataset showed **severe class imbalance**
- Some classes (e.g., melanocytic nevi) dominate the dataset
- Minority classes have significantly fewer samples

This imbalance can cause:

- Bias toward majority classes
- Poor detection of rare but critical cancers

3.2 Visualization

- Bar plots were generated to visualize:
    - Original class distribution
    - Balanced dataset distribution

3.3 Metadata Analysis

- Age distribution was analyzed
- Class frequency visualization confirmed imbalance

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4. Data Preprocessing

4.1 Data Cleaning

- Images were verified and filtered during loading
- Corrupted or unreadable images were skipped

4.2 Data Balancing

- Dataset balanced to **2000 images per class**

- Strategy:
    - Random sampling for large classes
    - Data augmentation for smaller classes

4.3 Data Augmentation

Using **Albumentations**, the following transformations were applied:

- Horizontal & vertical flips
- Rotation
- Shift, scale, rotate
- Brightness & contrast adjustment
- CLAHE (contrast enhancement)
- Gaussian noise

Purpose:

- Increase dataset diversity
- Reduce overfitting

4.4 Hair Removal (Noise Reduction)

A custom preprocessing function was implemented:

- Blackhat morphological operation
- Thresholding
- Inpainting

Purpose:

- Remove hair artifacts that may mislead the model

4.5 Lesion Segmentation

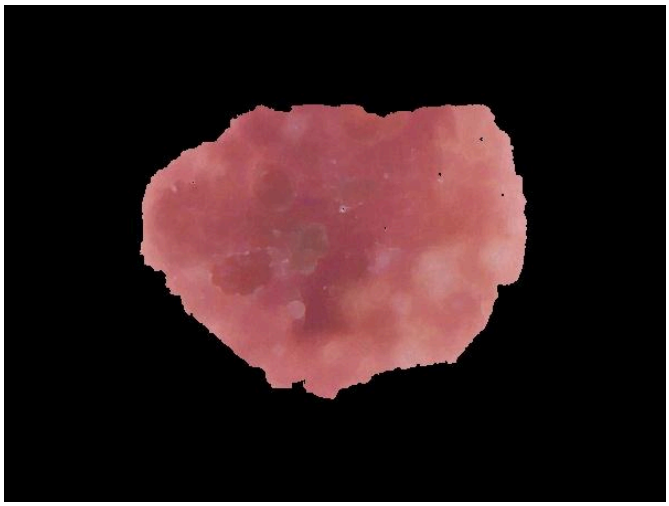
To further improve model focus on relevant regions, a lesion segmentation step was explored using classical computer vision techniques from **OpenCV**.

Method Used: GrabCut Algorithm

The segmentation approach was based on the **GrabCut algorithm**, which separates foreground (lesion) from background using iterative graph cuts.

Procedure

1. Initialize a mask for the image
2. Define a rectangular region assumed to contain the lesion
3. Apply GrabCut to iteratively refine foreground and background
4. Generate a binary mask
5. Extract the lesion by masking the original image.



After segmentation & hair removal

4.5 Dataset Splitting

- Training: 70%
- Validation: 15%
- Test: 15%

Split was done per class
Prevents data leakage

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5. Model Architecture

5.1 Base Model

- Pretrained **ResNet50**
- ImageNet weights used for feature extraction

5.2 Custom Classification Head

Added layers:

- Global Average Pooling
- Batch Normalization
- Dense (512 → 256 → 128)
- Dropout (0.5, 0.3)
- Softmax output layer

5.3 Loss Function

- Categorical Crossentropy
- Label smoothing (0.1)

5.4 Class Imbalance Handling

- Class weights computed dynamically using:
    - `compute_class_weight()`

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6. Training Strategy

Phase 1: Feature Extraction

- Base model frozen
- Only top layers trained

Phase 2: Fine-Tuning

- Base model unfrozen partially
- First 100 layers frozen
- Remaining layers trained

Training Configuration

- Optimizer: Adam
- Learning rate:
    - Phase 1: 1e-4
    - Phase 2: 1e-5
- Epochs: up to 50
- Batch size: 32

Regularization Techniques

- Early Stopping
- ReduceLROnPlateau
- Model Checkpointing

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7. Model Evaluation

7.1 Metrics Used

- Accuracy
- Classification Report (Precision, Recall, F1-score)
- Confusion Matrix

7.2 Results

- Validation accuracy computed after training
- Test accuracy evaluated on unseen data

```
best_model = tf.keras.models.load_model("best_model.keras")

loss, acc = best_model.evaluate(test_data)
print("Test Accuracy:", acc)
```

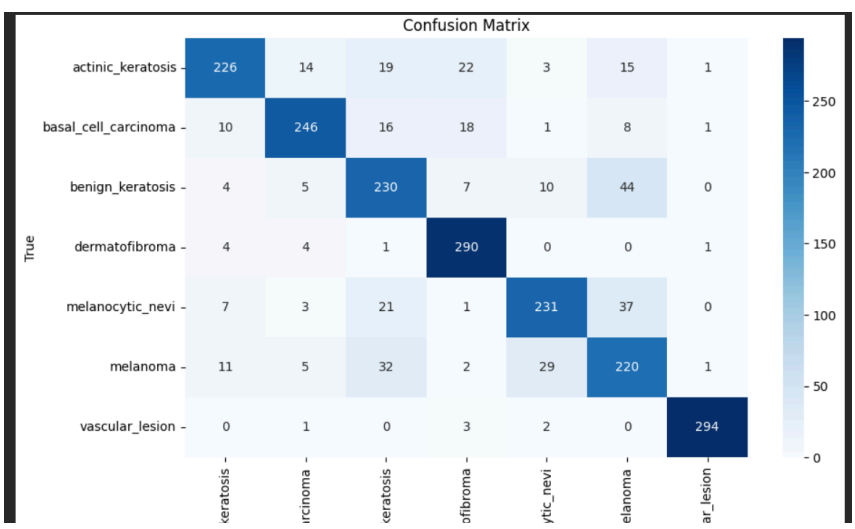
```
66/66 ━━━━━━━━━━━━━━━━━━━━ 22s 239ms/step - accuracy: 0.8229 - loss: 0.4844
Test Accuracy: 0.822857141494751
```

```
66/66 ━━━━━━━━━━━━━━━━━━━━ 27s 260ms/step
                     precision    recall  f1-score   support

   actinic_keratosis       0.86      0.75      0.80       300
basal_cell_carcinoma       0.88      0.82      0.85       300
    benign_keratosis       0.72      0.77      0.74       300
      dermatofibroma       0.85      0.97      0.90       300
    melanocytic_nevi       0.84      0.77      0.80       300
            melanoma       0.68      0.73      0.71       300
     vascular_lesion       0.99      0.98      0.98       300

            accuracy                           0.83      2100
           macro avg       0.83      0.83      0.83      2100
        weighted avg       0.83      0.83      0.83      2100
```

Before i add segmentation part

7.3 Confusion Matrix

- Visualized using heatmap
- Shows per-class prediction performance



But this before segmentation

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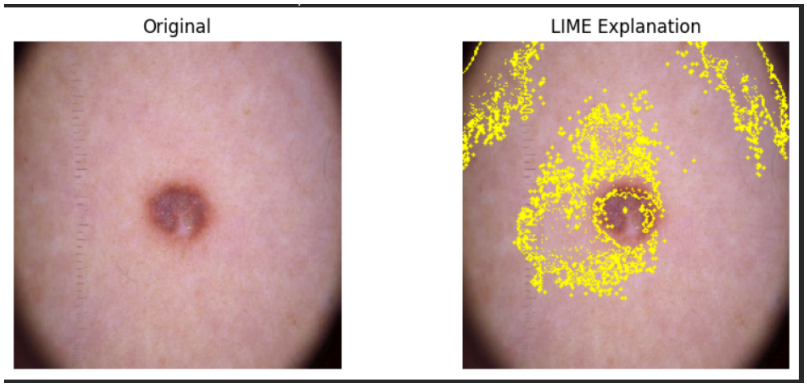
8. Explainable AI (XAI)

8.1 Techniques Used

- Grad-CAM (planned) but Colab crashed many times
- LIME



This is before segmentation too i think the model was training on background too so i added segmentation part

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8.2 Purpose

- Understand model decisions
- Ensure focus on lesion regions

8.3 Interpretation

- Heatmaps highlight important regions
- Helps verify:
    - Model is learning relevant medical features
    - Not relying on background noise

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9. Challenges

9.1 Class Imbalance

- Required aggressive augmentation and class weighting

9.2 Computational Constraints

- Large dataset + deep model
- Memory limitations required optimization

9.3 Model Generalization

- Risk of overfitting due to dataset duplication

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10. Future Work

- Implement segmentation models (e.g., U-Net)
- Improve rare class detection
- Compare with:
    - EfficientNet
    - Inception models
- Expand XAI analysis
- Deploy as a clinical decision support tool

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11. Conclusion

This project successfully developed a complete deep learning pipeline for skin cancer classification. By combining:

- Advanced preprocessing

- Transfer learning
- Class balancing
- Explainable AI

References

Khan, T., Haque, M. Z. U., Munir, G., & Usmani, I. A. (2026). Deep Learning-Based Skin Care Detection with Multi-method Explainability: Grad-CAM, Lime, and Occlusion Sensitivity. *IIUM Engineering Journal*, *27*(1), 160–174. https://doi.org/10.31436/iiumej.v27i1.4049

Hasan, M. K., Elahi, M. T. E., Alam, M. A., Jawad, M. T., & Martí, R. (2022). DermoExpert: Skin lesion classification using a hybrid convolutional neural network through segmentation, transfer learning, and augmentation. *Informatics in Medicine Unlocked*, *28*, 100819. https://doi.org/10.1016/j.imu.2021.100819

Dagnaw, G. H., Mouhtadi, M. E., & Mustapha, M. (2024). Skin cancer classification using vision transformers and explainable artificial intelligence. *Journal of Medical Artificial Intelligence*, *7*, 14. https://doi.org/10.21037/jmai-24-6